

Name: _____

Date: _____

SOLUTIONS



Year 8 Mathematics

Diagnostic Test

CALCULATOR FREE

1. Write the value of the 7 in each number.

a 1783

seven hundred (700)

b 71 953

seventy thousand (70 000)

c 10 073

seventy (70)

d 107 583 928

seven million (7 000 000)

2. Write each number in words.

a 599

five hundred and ninety nine

b 2847

two thousand eight hundred and forty seven

c 80 132

eighty thousand one hundred and thirty two

3. Write the number:

a six thousand, eight hundred and thirty-four

6834

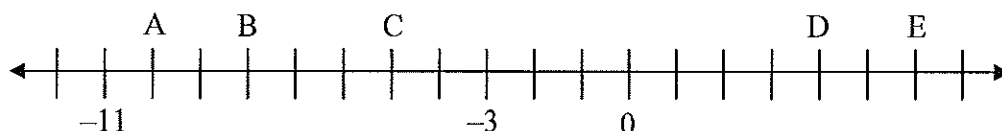
b nine hundred and twenty-seven thousand, five hundred and eighty-six

927 586

4. Complete the place value chart by writing the digits in the appropriate columns.

	Number	Place value parts							
		Tens of millions	Millions	Hundreds of thousands	Tens of thousands	Thousands	Hundreds	Tens	Ones
a	1904					1	9	0	4
b	42 085				4	2	0	8	5
c	1 392 657		1	3	9	2	6	5	7

5. Each letter on this number line represents a number. Write the numbers.

A is -10B is -8C is -5D is 4E is 6

6. Order these integers from smallest to largest.

a 51, 13, 5, 22, 6, 4, 15

4, 5, 6, 13, 15, 22, 51

b -18, 23, 345, -47, -99, -7, 17

-99, -47, -18, -7, 17, 23, 345

7. Write the integer that is:

a 5 smaller than 11

6

b 4 greater than 9

13

c 6 smaller than 200

194

d 8 greater than 55

63

8. Write the next four numbers in these sequences.

a 57, 50, 43, 36, 29, 22, 15, 8

b 2, 4, 8, 16, 32, 64, 128, 256

9. Which number makes a total of 86 when added to:

a 36?

50

b 75?

11

c 17?

69

10. Which number makes a total of 400 when added to:

a 354?

46

b 278?

122

c 62?

338

11. Calculate the following:

a $643 + 5176 + 98 + 882 =$ 6799

b $4375 + 725 + 241 + 67 =$ 5408

12. Solve these subtractions.

a $83 - 48 =$ 35

b $734 - 256 =$ 478

c $502 - 319 =$ 183

13. In year 6, there are 27 girls and 25 boys. 18 of these students live on the south side of the school, the rest live on the north side. How many live on the north side of the school?

34

14. Frank saved \$960 over six months. In the first two months he saved \$424. He saved a further \$295 in the next two months. How much did he save in the last two months?

\$241

15. a $4762 + 2519 =$

7281

b $3474 - 1656 =$

1818

16. a $16\ 845 + 42\ 076 =$

58921

b $75\ 305 - 28\ 647 =$

46658

17. a $487\ 086 + 788\ 935 =$

1276021

b $5\ 625\ 437 - 1\ 709\ 748 =$

3915689

18. Here are the approximate distances between the capital cities of mainland Australia:

Melbourne to Sydney 875 km

Darwin to Perth

4049 km

Sydney to Brisbane 1000 km

Perth to Adelaide

2724 km

Brisbane to Darwin 3394 km

Adelaide to Melbourne

904 km

- a Jan travelled from Perth to Adelaide. Peter travelled from Adelaide to Brisbane via Melbourne and Sydney. Calculate how far each person travelled. Who travelled the furthest?

Peter = 2779 km ∴ Peter went further

- b What is the longest journey between three connected capital cities?

Brisbane-Darwin-Perth = 7443 km

- c What is the shortest journey between three connected capital cities?

Melbourne-Sydney-Brisbane = 1875 km.

19. A charity raised the following amounts of money in four different locations.

\$456 287

\$919 706

\$202 439

\$866 731

Their target was \$3 million dollars.

- a Did they reach their target?

\$2445168 ∴ NO

- b How much more money will they need to reach their target?

\$554837

20. Write the first 10 multiples of 7.

7, 14, 21, 28, 35, 42, 49, 56, 63, 70

21. Write True or False to complete these statements.

a 18 is a multiple of 5 F

b 36 is a multiple of 3 T

c 24 is a multiple of 8 T

22. Circle the numbers that are multiples of 3.

66 21 67 12 13 23 36 42 20

23. There are six rows of eight chairs. How many chairs are there?

48

24. a The product of 8 and 5 is 40.

b 32 is the product of 4 and 8. (2 and 16)

c 77 is the product of 7 and 11.

25. Write True or False to complete these statements.

a 2 is a factor of 17 F

b 3 is a factor of 35 F

c 9 is a factor of 63 T

26. List the factors of:

a 36 1, 2, 3, 4, 6, 9, 12, 18, 36

b 50 1, 2, 5, 10, 25, 50

27. Circle the numbers that have 7 as a factor.

21 9 77 24 74 63 14 84 37 70

28. Circle the number in each line that does not belong. Explain your reason for excluding the number.

a 24 20 8 6 12 16
Not a multiple of 4

b 66 61 63 69 71 65
Not an odd number

c 13 16 19 17 23 11
Not a prime number

29. List the prime numbers between 0 and 10. 2, 3, 5, 7

30. List the composite numbers between 10 and 20. 12, 14, 15, 16, 18

31. Fill the gaps to complete these multiplications.

$$\begin{aligned} \text{a } 5 \times 17 &= 5 \times (10 + \underline{7}) \\ &= 5 \times \underline{10} + 5 \times \underline{7} \\ &= \underline{50} + \underline{35} \\ &= \underline{85} \end{aligned}$$

$$\begin{aligned} \text{b } 20 \times 35 &= 20 \times (\underline{30} + 5) \\ &= 20 \times \underline{30} + \underline{20} \times 5 \\ &= \underline{600} + \underline{100} \\ &= \underline{700} \end{aligned}$$

32. Complete the following:

a 33×4 132

b 65×8 520

c 8217×10 82170

d 573×9 5157

e 387×11 4257

f 237×5 1185

g 429×20 8580

h 80×25 2000

33. James got this answer for the following multiplication:

$$72 \times 15 = 432$$

Was he correct? If not, what should his answer be?

NO
1080

34. Calculate:

a 84×36 3024

b 277×48 13296

c 391×46 17986

35. The attendance at the football was 6842 people at 12:30 pm. It was 6 times larger at 1:30 pm. It continued to increase by 1150 each minute from 1:30 until 2 pm. What was the final attendance figure?

$$\begin{aligned} 6 \times 6842 &= 41052 \\ 30 \times 1150 &= 34500 \\ \text{total} &= \underline{75552} \end{aligned}$$

36. Calculate:

a $832 \div 8$

104

b $649 \div 7$

92 r 5 or $92\frac{5}{7}$

c $584 \div 6$

97 r 1 or $97\frac{1}{3}$

d $4275 \div 5$

855

e $936 \div 4$

234

f $924 \div 14$

66

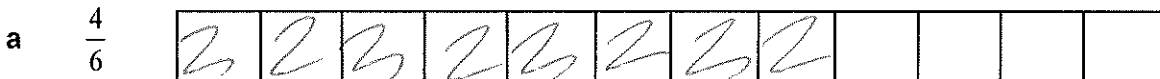
37. Complete this measurement conversion table. Use decimals if necessary.

	Millimetres	Centimetres	Metres	Kilometres
a	1000000 mm	10000 cm	1000 m	1 km
b	4000 mm	400 cm	4 m	0.004 km
c	920 mm	92 cm	0.92 m	0.00092 km
d	3800000 mm	380000 cm	3800 m	3.8 km

38. Write the fraction that corresponds to the shaded part of each football scarf.



39. Colour the scarf to show each fraction.



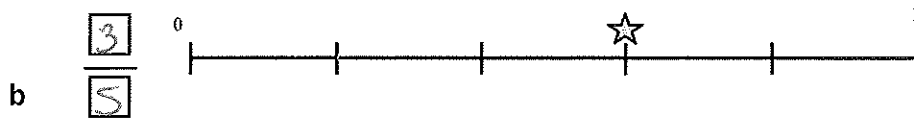
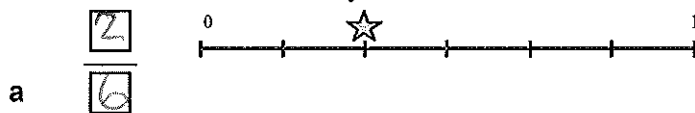
40. A bakery makes a mixed dozen selection of bread rolls. Of the 12 rolls, 2 are poppy seed, 2 are wholemeal, 3 are white, 4 are multigrain and one is cape seed.

a What fraction of the rolls are white? $\frac{3}{12} = \frac{1}{4}$

b What fraction of the rolls are wholemeal? $\frac{2}{12} = \frac{1}{6}$

c What fraction of the rolls are poppy seed or multigrain or cape seed? $\frac{7}{12}$

41. Write the fraction shown by the star on each number line.



42. Fill in the missing numerators and denominators to make equivalent fractions.

a $\frac{1}{\boxed{3}} = \frac{4}{12} = \frac{\boxed{5}}{15} = \frac{3}{\boxed{9}}$

b $\frac{3}{12} = \frac{\boxed{1}}{4} = \frac{4}{\boxed{16}} = \frac{5}{\boxed{20}}$

43. Order these fractions from smallest to largest.

a $\frac{3}{7}, \frac{8}{7}, \frac{1}{7}, \frac{4}{7}, \frac{6}{7}$ $\frac{1}{7}, \frac{3}{7}, \frac{4}{7}, \frac{6}{7}, \frac{8}{7}$

b $\frac{1}{30}, \frac{1}{20}, \frac{8}{20}, \frac{1}{10}, \frac{6}{10}$ $\frac{1}{30}, \frac{1}{20}, \frac{1}{10}, \frac{8}{20}, \frac{6}{10}$

c $\frac{3}{12}, \frac{1}{6}, \frac{23}{24}, \frac{3}{4}, \frac{1}{2}$ $\frac{1}{6}, \frac{3}{12}, \frac{1}{2}, \frac{3}{4}, \frac{23}{24}$

44. Write 'proper fraction', 'improper fraction' or 'mixed number' for each.

a $2\frac{4}{7}$ mixed

b $\frac{3}{4}$ proper

c $\frac{14}{15}$ proper

d $1\frac{14}{13}$ mixed

e $\frac{14}{8}$ improper

f $5\frac{12}{17}$ mixed.

45. Convert these mixed numbers to improper fractions.

a $5\frac{1}{3}$ $\frac{16}{3}$

b $7\frac{5}{8}$ $\frac{61}{8}$

c $3\frac{7}{9}$ $\frac{34}{9}$

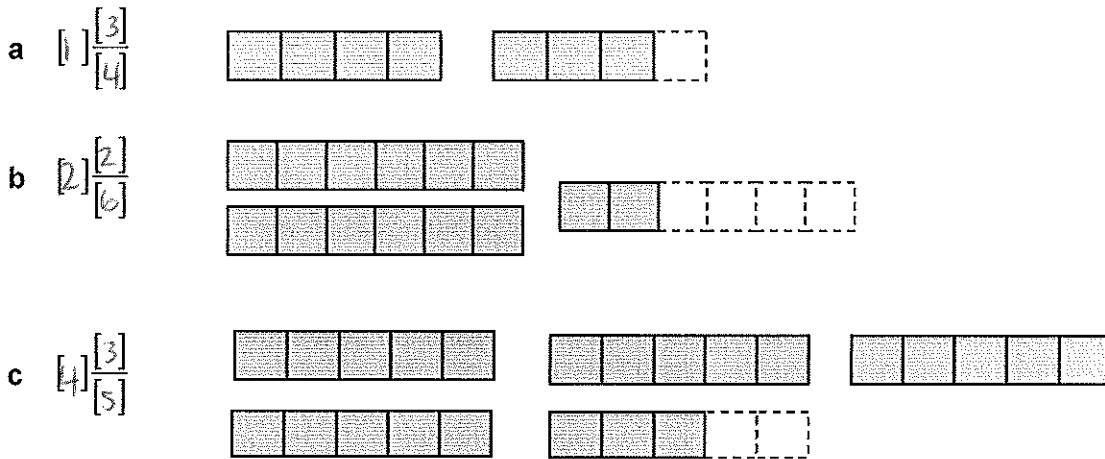
46. Convert these improper fractions to mixed numbers.

a $\frac{29}{5}$ $5\frac{4}{5}$

b $\frac{60}{7}$ $8\frac{4}{7}$

c $\frac{34}{8}$ $4\frac{2}{8}$ or $4\frac{1}{4}$

47. Write the mixed number to describe how much of a chocolate bar is shaded.



48. The distance Manny walks to school is $1\frac{1}{3}$ kilometres each way. How far does he walk in:

a one day? $2\frac{2}{3}$ km

b two days? $5\frac{1}{3}$ km

c three days? 8 km

d five days? $13\frac{1}{3}$ km

49. Add these fractions. Write your answers as proper fractions or mixed numbers.

a $\frac{1}{8} + \frac{4}{8} + \frac{2}{8} = \frac{7}{8}$

b $\frac{1}{5} + \frac{2}{5} + \frac{4}{5} = \frac{7}{5} = 1\frac{2}{5}$

c $\frac{8}{20} + \frac{4}{20} + \frac{1}{20} + \frac{5}{20} + \frac{6}{20} = \frac{24}{20} = 1\frac{1}{5}$

50. Circle the correct answer.

a $\frac{2}{7} + \frac{2}{7} = \frac{4}{14}$ or $\left(\frac{4}{7}\right)$ or $\frac{22}{7}$

b $\frac{5}{18} + \frac{11}{18} = \left(\frac{16}{18}\right)$ or $\frac{16}{36}$ or $\frac{6}{18}$

51. Pat wrote down how much petrol his Dad used on each day for one week.

	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
Fraction of a litre of petrol used	$\frac{6}{4}$	$\frac{5}{4}$	$\frac{2}{4}$	$\frac{1}{4}$	$\frac{11}{4}$	$\frac{7}{4}$	$\frac{9}{4}$

- a On which days was more than one litre of petrol used?

Sat, Sun, Wed, Thur, Fri

- b How much petrol was used from Monday to Friday? $\frac{30}{4} = 7\frac{2}{4} = 7\frac{1}{2} L$

- c How much petrol was used on the weekend? $\frac{11}{4} = 2\frac{3}{4} L$

- d How much petrol was used in total for the week? $\frac{41}{4} = 10\frac{1}{4} L$

52. Add the fractions.

a $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$

b $\frac{1}{3} + \frac{1}{5} = \frac{8}{15}$

53. Add these fractions. Write the answer as a mixed number.

a $\frac{5}{7} + \frac{1}{3} = 1\frac{1}{21}$

b $\frac{5}{6} + \frac{3}{4} = 1\frac{7}{12}$

c $\frac{1}{3} + \frac{4}{6} + \frac{7}{9} + \frac{5}{12} = 2\frac{7}{36}$

54. Work out these subtractions.

a $\frac{11}{12} - \frac{5}{12} = \frac{6}{12} = \frac{1}{2}$

b $\frac{4}{6} - \frac{3}{18} = \frac{9}{18} = \frac{1}{2}$

c $\frac{2}{3} - \frac{3}{5} = \frac{1}{15}$

55. The River family drove home from their holiday in Fingal Bay. It took them four days. On the first day they drove $\frac{1}{3}$ of the distance. On the second day they drove $\frac{1}{4}$ of the distance. At the end of the third day, they still had $\frac{1}{4}$ of their trip to go. How far did they drive on day three?

$\frac{3}{4} - \frac{1}{3} - \frac{1}{4} = \frac{1}{6}$ of the journey

56. a Calculate $\frac{3}{4}$ of 24. 18
- b Calculate $\frac{5}{9}$ of 27. 15
- c Calculate $\frac{7}{10}$ of 150. 105

57. a $\frac{1}{2}$ of $\frac{1}{10}$ = $\frac{1}{2} \times \frac{1}{10} = \frac{1}{20}$
- b $\frac{1}{5}$ of $\frac{1}{3}$ = $\frac{1}{5} \times \frac{1}{3} = \frac{1}{15}$
- c $\frac{1}{4}$ of $\frac{1}{8}$ = $\frac{1}{4} \times \frac{1}{8} = \frac{1}{32}$

58. Find the value of each division. Write your answer as a fraction or a mixed number in simplest form.

a $5 \div 9 = \frac{5}{9}$

b $18 \div 20 = \frac{18}{20} = \frac{9}{10}$

c $20 \div 13 = \frac{20}{13} = 1 \frac{7}{13}$

d $30 \div 8 = \frac{30}{8} = 3 \frac{3}{4}$

59. Margaret sold 75 raffle tickets. Each raffle book had 12 tickets. How many raffle books did she sell? Give your answer as a mixed number.

$6 \frac{1}{4}$

60. Fill in the blanks to show the place value of each decimal.

a $2.6 = \boxed{2} + \frac{\boxed{6}}{10}$

b $7.45 = \boxed{7} + \frac{\boxed{4}}{10} + \frac{\boxed{5}}{100}$

c $30.07 = \boxed{30} + \frac{\boxed{0}}{10} + \frac{\boxed{7}}{100} = \boxed{30} + \frac{\boxed{7}}{100}$

61. Fill in the blanks.

- a When the number line is marked in tenths,
9.37 is between 9. 3 and 9. 4.

- b When the number line is marked in hundredths,
19.685 is between 19. 6 8 and 19. 6 9.

62. a $592.64 = \boxed{5} \boxed{9} \boxed{2} + \frac{\boxed{6}}{10} + \frac{\boxed{4}}{100}$

b $673.09 = \boxed{6} \boxed{7} \boxed{3} + \frac{\boxed{0}}{10} + \frac{\boxed{9}}{100}$

63. Write these fractions and mixed numbers as decimals.

a $18\frac{4}{10} = 18.4$

b $33\frac{15}{100} = 33.15$

c $\frac{3}{10} = 0.3$

d $61\frac{129}{100} = 61 + 1\frac{29}{100} = 62.29$

64. Write these decimals as mixed numbers or fractions.

a $0.7 = \frac{7}{10}$

b $0.94 = \frac{94}{100} = \frac{47}{50}$

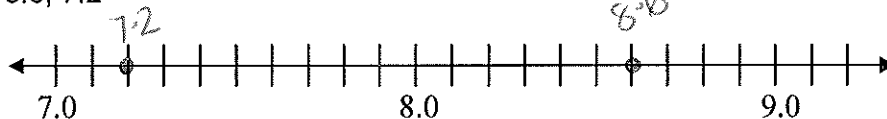
c $2.75 = 2\frac{75}{100} = 2\frac{3}{4}$

d $11.01 = 11\frac{1}{100}$

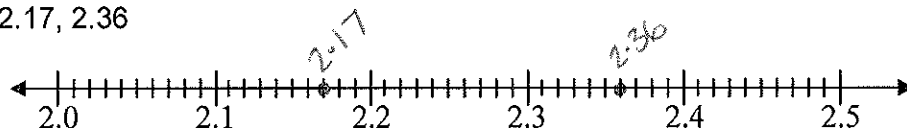
e $121.985 = 121\frac{985}{1000} = 121\frac{197}{200}$

65. Mark the numbers on the number line.

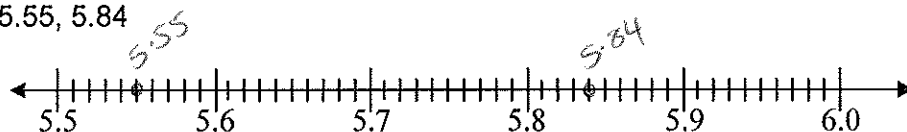
a 8.6, 7.2



b 2.17, 2.36



c 5.55, 5.84



66. Arrange the numbers from smallest to largest.

a 13.1, 13.8, 13.4, 13.2, 13.5, 13.6, 13.3, 13.7

13.1, 13.2, 13.3, 13.4, 13.5, 13.6, 13.7, 13.8

b 25.94, 25.49, 25.98, 25.89, 25.70, 25.07

25.07, 25.49, 25.70, 25.89, 25.94, 25.98

c 0.05, 0.802, 0.9, 0.4571, 0.007, 0.63

0.007, 0.05, 0.4571, 0.63, 0.802, 0.9

67. Jack took some pencils out of his pencil case and measured them. He discovered later that each pencil was actually $\frac{1}{10}$ cm shorter than what he measured.
- What are the real lengths of each pencil? Write them in the table below. Then write each measure as a fraction.

		Jack's measure	Make each one $\frac{1}{10}$ cm shorter	Write as a fraction
a	Pencil 1	11.6 cm	11.5 cm	$11\frac{5}{10} = 11\frac{1}{2}$ cm
b	Pencil 2	20.7 cm	20.6 cm	$20\frac{6}{10} = 20\frac{3}{5}$ cm
c	Pencil 3	9.85 cm	9.75 cm	$9\frac{75}{100} = 9\frac{3}{4}$ cm
d	Pencil 4	20.65 cm	20.55 cm	$20\frac{55}{100} = 20\frac{11}{20}$ cm
e	Pencil 5	14.18 cm	14.08 cm	$14\frac{8}{100} = 14\frac{2}{25}$ cm

68. Convert these decimals to mixed numbers or fractions.

a $11.85 = 11\frac{85}{100} = 11\frac{17}{20}$ b $46.02 = 46\frac{2}{100} = 46\frac{1}{50}$

c $176.102 = 176\frac{102}{1000} = 176\frac{51}{500}$ d $592.011 = 592\frac{11}{1000}$

69. Convert these mixed numbers to decimals.

a $2\frac{7}{10} = 2.7$ b $12\frac{14}{100} = 12.14$

c $37\frac{3}{100} = 37.03$ d $28\frac{645}{1000} = 28.645$

e $91\frac{75}{1000} = 91.075$ f $73\frac{6}{1000} = 73.006$

70. Write an equivalent fraction with denominator of 10 for each of these fractions, and then write each of the fractions as a decimal.

a $\frac{1}{5} = \frac{2}{10} = 0.2$

b $\frac{3}{5} = \frac{6}{10} = 0.6$

71. Write each of these as an equivalent fraction with denominator of 100 and then write each fraction as a decimal.

a $\frac{1}{2} = \frac{50}{100} = 0.5$

b $\frac{2}{25} = \frac{8}{100} = 0.08$

72. Write these mixed numbers as decimals.

a $7\frac{1}{2} = 7.5$

c $3\frac{6}{50} = 3\frac{12}{100} = 3.12$

73. Calculate:

a $17.2 + 2.4 = 19.6$

b $93.42 + 4.36 = 97.78$

c $7.04 + 1.76 = 8.8$

d $7.425 + 0.207 + 73.47 + 883.5 = 964.602$

74. Use one of the subtraction algorithms to calculate:

a $9.6 - 2.4 = 7.2$

b $12.36 - 9.14 = 3.22$

c $17.2 - 4.5 = 12.7$

d $24.34 - 8.76 = 15.58$

75. Josh earned the following amounts of money for one week.

	Monday	Tuesday	Wednesday	Thursday	Friday
Earnings	\$18.45	\$33.78	\$18.13	\$15.84	\$27.64

a At the end of which day could he afford to buy a game costing \$60.00?

Wednesday

b How much will he put in the bank at the end of the week, after he pays for his game and keeps \$20.00 cash?

$113.84 - 60 - 20 = \$33.84$

76. Complete the table by multiplying each number by 10, 100 and 1000.

	Number	$\times 10$	$\times 100$	$\times 1000$
a	2.2	22	220	2200
b	97.8	978	9780	97800
c	0.002	0.02	0.2	2
d	11.036	110.36	1103.6	11036
e	124.204	1242.04	12420.4	124204
f	80.093	800.93	8009.3	80093

77. Divide each of these numbers by 10:

a $197 =$ 19.7

b $8.6 =$ 0.86

c $482.74 =$ 48.274

78. Calculate:

a $1.8 \div 6$ 0.3

b $4.5 \div 9$ 0.5

c $5.34 \div 3$ 1.78

79. Calculate 0.6 of:

a \$20 \$12

b 25 grams 15 grams

c 980 metres 588 m

d \$245 \$147

80. Round these numbers to the nearest ten.

a 23 23

b 48 50

c 99 100

d 1008 1010

81. Round these numbers to the nearest whole number.

a 45.2 45

b 37.8 38

c 82.56 83

d 27.333 27

82. Complete the table.

		Rounded to the nearest whole number	Rounded to the nearest tenth	Rounded to the nearest hundredth
a	2.345	2	2.3	2.35
b	11.83726	12	11.8	11.84
c	498.00234	498	498.0	498.00

83. Write the percentage for each situation.

- a 46 out of 100 people went on a holiday.

46%

- b 3 out of 10 horses were ready to race.

30%

84. Write each percentage as a fraction or a mixed number.

- a 14%

$$\frac{14}{100} = \frac{7}{50}$$

- b 87%

$$\frac{87}{100}$$

- c 319%

$$3\frac{19}{100}$$

85. Write these fractions as percentages.

a $\frac{18}{100}$

18%

b $\frac{71}{100}$

71%

c $\frac{6}{50}$

12%

d $\frac{8}{25}$

32%

86. Complete the table with the corresponding fractions and mixed numbers, decimals and percentages.

	Decimal	Fraction / mixed number	Percentage
a	0.39	$\frac{39}{100}$	39%
b	0.08	$\frac{8}{100} = \frac{2}{25}$	8%
c	1.45	$1\frac{45}{100}$	145%
d	7.03	$7\frac{3}{100}$	703%

87. In the Cranbourne Botanic Gardens, 57% of the trees are native to the region. What percentage of the trees are not native?

$$100 - 57 = 43\%$$

88. Calculate these numbers.

- a 3% of 100

3

- b 5% of 600

$$10\% = 60$$

30

- c 16% of 400

$$1\% = 4$$

64

- d 200% of 850

1700

89. Write these statements in words. Use the word 'something' in place of the pronumeral.

a $x - 9 = 17$

something subtract nine equals 17

b $13 + y = 21$

13 plus something equals 21

c $z \div 7 = 4$

something divided by 7 equals 4

d $8 \times r = 24$

8 times something equals 24

90. Find the value of the pronumeral.

a $14 + k = 23$

$k = 9$

b $m - 5 = 19$

$m = 24$

c $3 \times n = 33$

$n = 11$

d $p \div 6 = 5$

$p = 30$

e $q + 25 = 150$

$q = 125$

f $119 - r = 102$

$r = 17$

g $s \times 12 = 144$

$s = 12$

h $100 \div t = 25$

$t = 4$